SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Pecel Lele
Project / Solution Name	Adept
Name (Leader)	David Kurniawan
Email & Phone (Leader)	davidk.academic@gmail.com & +6281229700811
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://adept-my.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/pecel-lelele/adept

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Adept is publicly deployed on Replit using the sponsor-provided Core subscription and Replit integrated PostgreSQL. One public Next.js standalone server renders pages, runs server actions and forwards API requests to private FastAPI on loopback. The root Python launcher checks configuration and API/database readiness, supervises both children, and performs no startup migrations or seeding. HTTP-only signed cookies hold local account sessions; PostgreSQL stores profiles, memberships, plans, evidence and catalogue data.

The batch research pipeline calls a pinned 384-dimensional MiniLM encoder on RunPod and validates dated trend snapshots before atomic publication. Optional 768-dimensional Gemini embeddings map unfamiliar skill phrases to catalogue suggestions. Matching and pathways use deterministic rules. Adi can use GLM-5.3 with Gemini failover to select validated answer-block IDs; displayed wording comes from server-authored facts.

Public employer/remote-job feeds, sourced courses, certification fees, official UN SDGs and BNM exchange rates complement research signals. GitHub Actions runs quality checks, database recovery tests, the actual root deployment smoke, weekly source audits and conditional trend refresh. Local Compose provides PostgreSQL only. The public site's reachability and MiniLM inference were observed separately; latest-source deployment and successful new snapshot publication require their own verification.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python >=3.11 (deployment runtime 3.11)	API, pipelines and tooling	Python Software Foundation License
Node.js 24; pnpm 10.29.3	Frontend and API-proxy runtime; package management	MIT
Next.js 16.4.2; React and React DOM 19.2.9	Server rendering, routing and UI	MIT

Name & Version	Purpose in Project	License
TypeScript ~5.9.3	Strict typing	Apache-2.0
Tailwind CSS / @tailwindcss/postcss 4.1.18	Styling	MIT
next-intl 4.14.9	English, Bahasa Melayu and Chinese	MIT
Three.js 0.180.0	Adi's interactive 3D mascot	MIT
lucide-react ^1.61.0	Interface icons	ISC
class-variance-authority 0.7.1	Component variants	Apache-2.0
clsx 2.1.1; tailwind-merge 3.3.1	CSS class composition	MIT
@fontsource-variable/inter ^5.3.0	Installed font dependency; current global UI styling uses Arial	MIT package; SIL OFL 1.1 font
openapi-fetch 0.14.1; openapi-typescript 7.9.1	Typed API client and generated contracts	MIT
@types/node 22.20.5; @types/react 19.2.18; @types/react-dom 19.2.7; @types/three 0.180.0	Development type definitions	MIT
FastAPI ≥0.161.1	HTTP API and OpenAPI	MIT
Pydantic ≥2.7,<3; pydantic-settings ≥2.16.1	Contracts and configuration	MIT
Uvicorn ≥0.54.0	ASGI service	BSD-3-Clause
psycopg[binary] ≥3.3.6; psycopg-pool ≥3.3.3	PostgreSQL access and pooling	LGPL-3.0-only; bundled binary components have additional notices
python-multipart ≥0.0.32	Multipart CV uploads	Apache-2.0
Python standard library	Signed local sessions and scrypt password hashing	PSF-2.0
PyMuPDF ≥1.24.0,<2	In-memory PDF text extraction.	AGPL-3.0 or Artifex commercial licence
HTTPX ≥0.28.1	External API and model-provider requests	BSD-3-Clause

Name & Version	Purpose in Project	License
PostgreSQL 16; pgvector (pgvector/pgvector:pg16; extension version not pinned)	Relational storage and separate vector indexes for 384-dimensional MiniLM embeddings and 768-dimensional Gemini embeddings.	PostgreSQL License
sentence-transformers 6.2.0; Transformers 5.20.2; RunPod Python SDK 1.12.0	Pinned MiniLM serverless worker (offline embeddings)	Apache-2.0 for Sentence Transformers and Transformers; MIT for RunPod Python SDK.
PyTorch 2.14.1	MiniLM worker dependency	BSD-3-Clause
pytest ≥9.1.1; Ruff ≥0.17.2; ESLint 9.39.5; eslint-config-next 16.4.2	Tests, quality checks and linting	MIT
uv	Python environment tooling; version is environment-dependent	MIT or Apache-2.0
Mermaid 11.12.0	Documentation-only diagram generation	MIT
PyYAML 6.0.3; actionlint 1.7.7	Workflow parsing and publication-policy validation	MIT
Prettier ^3.9.10	Development formatting	MIT
Playwright renderer	Documentation-only diagram rendering	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
arXiv cs.AI RSS (and query API for cached research windows)	Offline research-attention ingestion; historical replay windows also cite the query API. Current ingest uses RSS; no arXiv call on a recommendation request.	Public/free; rate limits and terms apply
UN Statistics SDG Goal and Target APIs	Goal and target sync; scheduled source audit	Public/free
Microsoft Learn Catalog API and exam-pricing page	Course/module metadata, catalogue sync, availability checks, certification prices	Public/free
Coursera courses API	Curated slug/workload availability checks; no subscription or sponsorship inferred	Public endpoint; provider terms apply
Bank Negara Malaysia OpenAPI	Exchange rates for MYR price estimates	Public/free
Public job-posting feeds: Greenhouse, Lever, Ashby, Workday, Jibe, Himalayas, Jobicy, Remote OK, Remotive	Dated vacancy and skill evidence; these nine source families occur in the committed 23 October extract. SmartRecruiters is excluded from configured sources.	Public access; each source's terms apply
Optional keyed job APIs: Careerjet, Jooble, Web3.career	Connectors exist but are skipped in the committed fetch log	Authenticated; provider terms and credentials required
Cited provider HTTPS pages (courses, certifications)	Course facts, link audits, certification-page reachability	Public page access; not a single API
RunPod: GLM-5.3 for Adi; Serverless MiniLM batch embeddings	Separate endpoints and credentials: Adi selects validated answer blocks; MiniLM generates offline 384-dimensional research/catalogue vectors	Server API keys; sponsor resource usage declared in section 5; provider terms and charges apply

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (3.5 Flash Lite, 3.6 Flash, 3.1 Flash Lite; Embedding 001)	Optional Adi failover in that order; optional 768-dimensional skill suggestions using redacted short phrases and an existing catalogue index	Optional authenticated API; API key required. Provider terms, usage limits and charges apply.

The BLS baseline is committed data, not an external API called by the running recommendation service. Model weights may be downloaded from Hugging Face when preparing the offline encoder; that is a model acquisition step, not hosted inference.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
GitHub repository / Actions	Version control, quality CI, source audit, worker publication and conditional trend refresh. The first real refresh failed saving to private database host helium.	Public repository; GitHub-hosted Linux runners (ubuntu-latest).
Local Docker Compose	PostgreSQL/pgvector only; web/API run through root commands	Local Windows Docker Desktop; no cloud region
Replit Autoscale	Hosts the deployed public prototype at https://adept-my.replit.app; One public Next.js server, private FastAPI; health does not identify the Git revision.	Sponsor-provided Replit Core subscription; Autoscale publishing.
Replit integrated PostgreSQL	Managed application database for accounts, profiles, catalogue records, snapshots, plans, progress and organisation data.	Replit built-in managed PostgreSQL attached to the Core project.
RunPod Serverless / hosted inference	Deployed MiniLM worker for offline embeddings and separate GLM-5.3 endpoint for Adi. Successful inference does not establish successful new snapshot publication.	US$25 sponsor-provided credits; usage-based Serverless and hosted inference.
GitHub Container Registry (GHCR)	Stores/distributes the MiniLM worker container image	Public ghcr.io/pecel-lelele/adept-minilm image

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	License / Terms	Usage (pre-trained / fine-tuned / self-trained)
sentence-transformers/all-MiniLM-L6-v2	Hugging Face, run on a Runpod serverless worker	Apache-2.0	Pre-trained, no fine-tuning. Offline 384-dimensional sentence/skill embeddings and cosine matching, threshold 0.55.
GLM-5.3 (glm-5.3)	Runpod, model glm-5.3	Provider terms; team to review before signing	Pre-trained hosted inference. Selects validated application-authored answer-block IDs for Adi; model-generated prose is not displayed. No team fine-tuning.
Gemini 3.5 Flash Lite; Gemini 3.6 Flash; Gemini 3.1 Flash Lite	Google Gemini API	Provider terms	Optional Adi failover using separately configured credentials and the listed model sequence. No team fine-tuning.

Model & Version	Source / Provider	License / Terms	Usage (pre-trained / fine-tuned / self-trained)
Gemini Embedding 001	Google Gemini API	Proprietary; Gemini API terms	Optional catalogue indexing/short skill-phrase suggestions, 768 dimensions. A separate purpose from MiniLM research embeddings; user confirmation required.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Custom skill and role catalogue (140 skills, 31 roles, 28 targets)	Taxonomy seed	Team-curated / synthetic	Team curation; upstream references retained; not a complete O*NET/ESCO import	No
O*NET 31.0 occupation crosswalk	onetcenter.org	Public-derived / curated	CC BY 4.0 with stated exceptions; credit and modifications required	No

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
arXiv cs.AI research signal	RSS ingest and cached windows	Public	arXiv terms; metadata and paper content have different rights; no blanket CC0 claim	No learner data; raw feed metadata may contain public author names
BLS 2023–2033 labour baseline	Committed extract, BLS	Public / team-derived	US government data, cite BLS; US baseline, not a Malaysian vacancy forecast	No
Canonical skill exposure and historical snapshots	Seed data and fixtures	Curated / public-derived	Reviewer-assigned exposure estimates, not probabilities; demo snapshots replay Oct 2024 research windows	No
17 SDG goals and 169 targets	UN SDG API, seed	Public	UN data terms and attribution	No
61 role-target and 165 skill-target links	Mapping seed	Team-curated	Draft team interpretations with cited evidence; no UN endorsement	No
Course records (98) and prerequisite graph	Catalogue sources, seed	Public facts / curated	Providers retain their terms; prices and HRD Corp notes are dated estimates	No

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Job-posting extract (1,210 records, committed 23 October 2026)	Public job boards and employer endpoints	Public / curated	Each board's terms; public access is not an open-data licence	No learner data; postings may name public employers
Certification and exam-pricing extract (14 exams)	Microsoft, AWS, Google Cloud, IIBA, IAPP pages	Public / curated	Issuer site terms; dated prices	No
Exchange rates	BNM OpenAPI	Public	BNM terms and attribution	No
Regression personas, organisation members and CV fixtures	Fixtures, persona seed, organisation seed	Synthetic	Team-created evaluation and test data only	No real identities
Five synthetic demo CVs and complete intake profiles: Imran Hakim, Sofia Tan, Meera Anand, Adrian Lim and Nur Hana	Team-provided PDFs and demo CV	Synthetic	Team-provided demo material; additional intake settings are synthetic assumptions.	No real identities

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✅ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex	Repository inspection, debugging, resolving Greptile findings, code/configuration corrections, checks and documentation review.	Major debugging and review support
Replit Agent	Exploring the repository and understanding its structure.	Minor production setup
Google Antigravity agent	Exploring the repository and understanding its structure.	Minor repository exploration
Anthropic Claude Code	Produced most of the application code; also supported demo-video creation and pitch-deck design concepts, layout and presentation material.	Major; primary code implementation and demo/presentation work
Greptile	Reviewing pull requests and identifying issues for the team to assess and resolve.	Moderate automated review support

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members reviewed AI-generated code and text before merging. Technical claims in the proposal and this declaration were checked against dependency manifests, routers, scoring code, migrations and seed files. The project quality gate (pytest, Ruff, ESLint, strict TypeScript) is run and its actual outcome reported. Public claims link to primary sources. Each member remains responsible for the work they submit, and the team leader approves the final submission. Automated checks do not establish legal compliance or workforce outcomes.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✅Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Runpod	US$25 sponsor credits; GLM-5.3 inference endpoint and Serverless MiniLM worker	Adi's validated answer-block selection and offline public-research/catalogue embeddings. Worker inference succeeded; successful new trend-snapshot publication is a separate check..
Replit	Sponsor-provided Core subscription, cloud deployment and integrated PostgreSQL	Hosts the public Adept web/API and managed application database. Replit Agent was also used for repository exploration, as declared in section 4.3.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit Core provides public hosting and integrated PostgreSQL. RunPod credits support MiniLM Serverless embeddings and GLM-5.3 inference for Adi. These services are replaceable without changing the core methods: the web/API can move to another host and compatible PostgreSQL service, MiniLM can run on other compute, and Adi supports optional Gemini failover plus a deterministic guide. Matching and pathway generation use deterministic rules and do not require generative inference. Existing trend snapshots remain usable when an offline refresh or model provider is unavailable.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demonstration uses synthetic identities, organisations and CVs. Replit integrated PostgreSQL stores account details, scrypt password hashes, skills, preferences, memberships, plans, progress and evidence links. CVs are parsed in memory; original files are not retained, and users confirm extracted skills before saving.

Optional Gemini embedding calls receive short redacted skill phrases. Adi's inference provider receives sanitised conversation text, workspace context and candidate answer blocks, which may contain personalised learning facts. Sanitisation does not guarantee complete anonymisation. MiniLM processes public research and catalogue text.

Signed HTTP-only session cookies, authorisation, sharing controls, upload limits and rate limits protect access. Certificate-link checks contact external provider pages. Account export and password-confirmed deletion are available.

Records remain until deliberately deleted; no automatic retention schedule is implemented, and immediate removal from backups is not guaranteed. Production secrets are configured outside the repository.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Research attention is a proxy for skill demand; US labour projections do not establish Malaysian vacancies. The small curated catalogue limits coverage. Seed snapshots contain historical replay data, so publication date and observed period must both be shown. Scenario and exposure scores are heuristic estimates, not probabilities or calibrated confidence intervals. SDG links are interpretations pending review, not verified social outcomes.

PDF skill extraction can miss aliases and scanned documents; users confirm skills. Provider page reachability does not prove certificate ownership. Prices, funding eligibility and API availability can change. Mitigations include dated sources, validation before snapshot publication, catalogue-ID checks, counterfactual tests, manual evidence review and estimate labels. Adi selects validated application-authored answer blocks and uses a deterministic guide when provider output fails

validation. Answer relevance still requires human judgment. Public deployment and automated checks do not establish security certification, fairness across all populations or improved employment outcomes.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

✅ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

✅ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

✅ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

✅ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

✅ Any use of official SDC sponsor resources has been declared in Section 5.

✅ The prototype described here is accessible to judges through the links provided in Section 1.

✅ All fields in this template have been completed within the word limits stated.

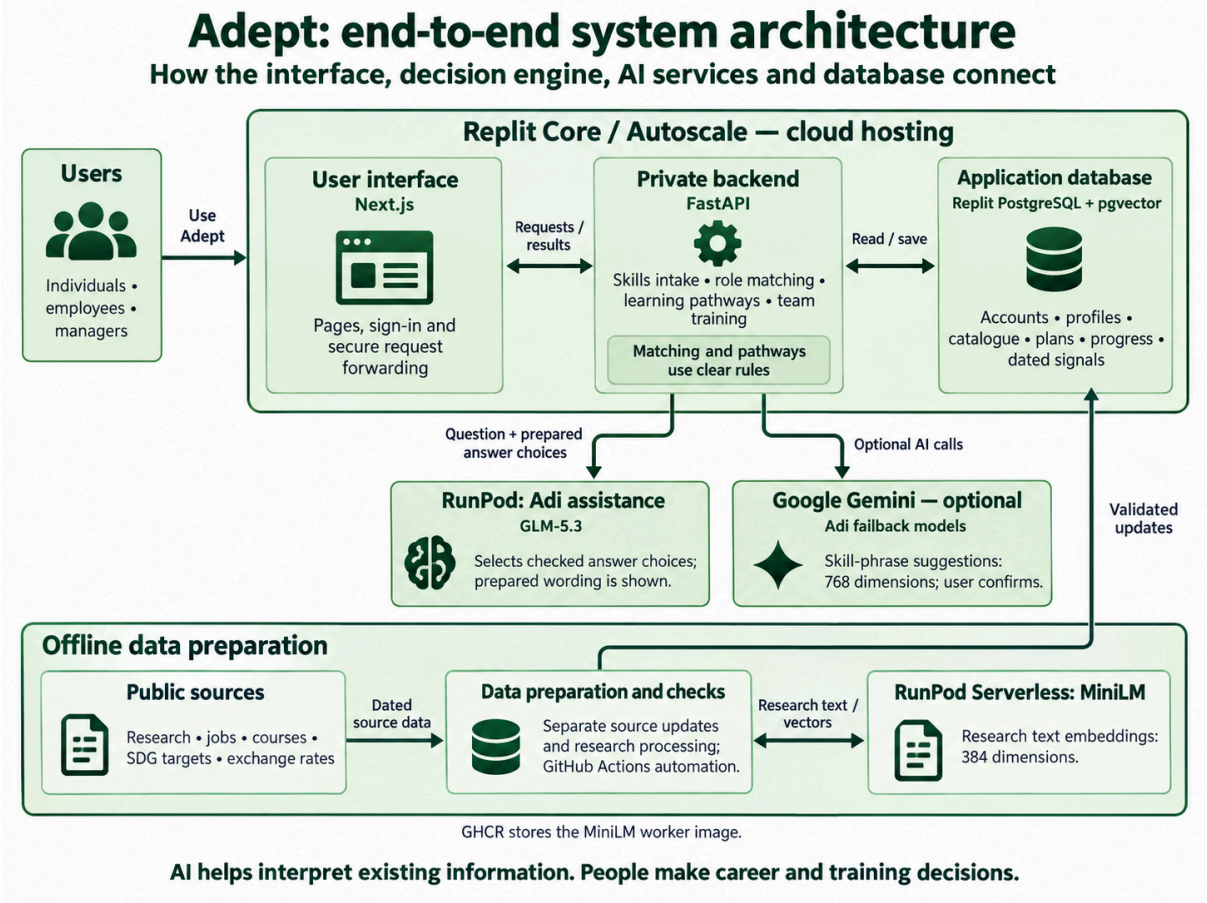
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Date] : 25/10/2026

Appendix


Adept: end-to-end system architecture
How the interface, decision engine, AI services and database connect
Replit Core / Autoscale — cloud hosting
Users
Individuals • employees • managers
Use Adept
User interface
Next.js
Pages, sign-in and secure request forwarding
Requests / results
Private backend
FastAPI
Skills intake • role matching • learning pathways • team training
Matching and pathways use clear rules
Read / save
Application database
Replit PostgreSQL + pgvector
Accounts • profiles • catalogue • plans • progress • dated signals
Question + prepared answer choices
Optional AI calls
RunPod: Adi assistance
GLM-5.3
Selects checked answer choices; prepared wording is shown.
Google Gemini — optional
Adi failback models
Skill-phrase suggestions: 768 dimensions; user confirms.
Validated updates
Offline data preparation
Public sources
Research • jobs • courses • SDG targets • exchange rates
Dated source data
Data preparation and checks
Separate source updates and research processing; GitHub Actions automation.
Research text / vectors
RunPod Serverless: MiniLM
Research text embeddings: 384 dimensions.
GHCR stores the MiniLM worker image.
AI helps interpret existing information. People make career and training decisions.